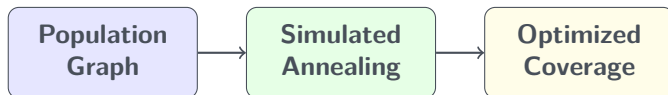


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# Hierarchical Healthcare Facility Location

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- **Integrated Network Design:** Strategically placing healthcare facilities where high-level hubs and local clinics are co-located to minimize infrastructure costs.
- **Graph-Based Optimization:** Modeling the region as a weighted graph—where nodes represent urban centers weighted by population—to determine the optimal placement of Tier-1 and Tier-2 facilities for maximum coverage with a limited budget.



- A **Decision Support System** for shifting from fragmented local development to a **coordinated strategic framework**.
  - *Emerging Contexts*: Designing optimized networks from the ground up.
  - *Legacy Contexts*: Rationalizing and reallocating hubs in over-saturated or inefficient areas.
- We exploit **economies of scope** through hierarchical nesting to minimize infrastructure costs, enabling **co-location savings**.
  - A generalizable framework applicable to any service network requiring tiered distribution (e.g., logistics, retail, or emergency services).

## The Hierarchical Facility Location Problem (HFLP)

- **Input:** A weighted graph  $G = (V, E)$ , where  $w_i$  represents the population of node  $i$ , and the number of facilities to be placed for each level  $(k_1, k_2)$ .
  - **Output:** An optimal allocation of facilities  $S_1, S_2 \subset V$ , such that  $|S_l| = k_l$  and  $S_2 \subseteq S_1$  (**Nested Hierarchy**)
  - **Objective:** Minimize  $Z = \sum_{j \in V} \sum_{l \in \text{Levels}} \alpha_l \cdot \Phi_j$ , where  $\Phi_j$  is a distance-based function to the nearest facility of level  $l$ .
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- The HFLP is Search Problem where the search space  $\mathcal{S}$  is composed by all possible permutation of facility locations. This solution space has exponential size:

$$|\mathcal{S}| = \binom{|V|}{k_1} \binom{k_1}{k_2}.$$

- Exact solvers (ILP) are computationally infeasible for large-scale networks due to the **combinatorial explosion** of the search space.

## Simulated Annealing

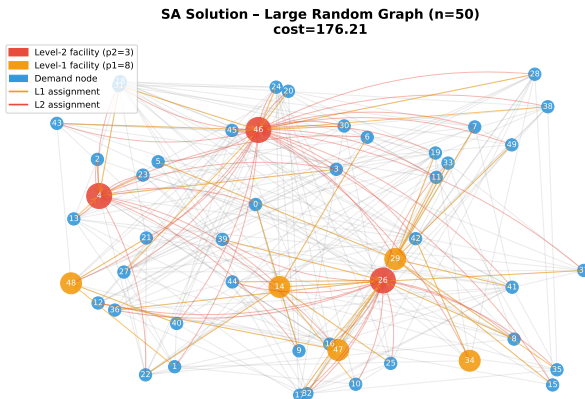
- **Probabilistic update:** Iteratively try some local changes that respect the constraints and accept each proposed change independently with probability:

$$P(S \rightarrow S') = \min \left\{ 1, e^{-\frac{\Delta Z}{T}} \right\}$$

**Temperature Schedule:** exponential  $T_{i+1} = \alpha T_i$ , with a fixed  $0 < \alpha < 1$ , or maybe something fancier.

- **Goal:** We aim to obtain near-optimal solutions in polynomial time, avoiding the exponential explosion of exact ILP solvers.
- **Validation:** Benchmarking on small synthetic networks, for which an exact solution is computable, to quantify the optimality gap.
- **Veneto Case Study:** We will apply our solver to the regional healthcare network to compare the optimized model with the current actual distribution.

- Preliminary results look promising: with small-size instances we get very small gap between ILP and SA solutions.
- Further testing on larger graphs and real-world cases is the next step.



## Questions?

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